

Emergent Specialization in Learner Populations: Competition as the Source of Diversity

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Abstract

How can populations of learners develop coordinated, diverse behaviors without explicit communication or diversity incentives? We demonstrate that **competition alone is sufficient** to induce emergent specialization—learners spontaneously partition into specialists for different environmental regimes through competitive dynamics, consistent with ecological niche theory. We introduce the **NichePopulation** algorithm, a simple mechanism combining competitive exclusion with niche affinity tracking via the canonical exponentiated-gradient (Hedge / multiplicative-weights) update on the regime simplex. Validated across **six real-world domains** (cryptocurrency trading, commodity prices, weather forecasting, solar irradiance, urban traffic, and air quality), our approach achieves a mean Specialization Index of **0.99** with effect sizes of Cohen’s $d > 50$. Key findings: (1) At $\lambda = 0$ (no niche bonus), learners still achieve mean SI ≈ 0.65 (and SI > 0.39 on every real-data domain), confirming specialization is genuinely emergent; (2) Diverse populations outperform homogeneous baselines by orders of magnitude in SI through method-level division of labor; (3) Our approach inherits the canonical $O(\sqrt{T \log R})$ no-regret guarantee of the Hedge algorithm and outperforms MARL baselines (QMIX, MAPPO, IQL) in specialization while being 4× faster. Code and data available at: <https://github.com/HowardLiYH/NichePopulation>.

1 Introduction

Coordinating populations of learners remains one of the fundamental challenges in machine learning. When multiple learners operate in shared environments, they face a critical dilemma: how to divide labor and specialize without explicit communication. This question connects to a deeper debate about the nature of intelligence itself: recent discussions among AI leaders have reignited questions about whether intelligence is inherently “general” or fundamentally specialized. LeCun has argued that human intelligence is “ridiculously specialized”—we can only compute an infinitesimal fraction of possible functions, and our sense of generality is an illusion LeCun (2022). Others maintain that brains are general in the Turing-complete sense, though practical systems inevitably require some degree of specialization. Our work contributes empirical evidence to this debate: we show that **specialization emerges inevitably from competition**, suggesting that the tendency to specialize is itself a universal property of bounded intelligent systems.

This challenge manifests across numerous domains with significant real-world impact. In autonomous driving, vehicles must implicitly coordinate traffic flow to avoid congestion and collisions. In algorithmic trading, strategies must avoid correlated behaviors that amplify market volatility—the 2010 Flash Crash, where the Dow Jones dropped 1,000 points in minutes, exemplifies the catastrophic consequences of homogeneous learner behaviors Kirilenko et al. (2017). In distributed sensing networks, sensors must specialize to different environmental conditions to maximize information gain.

Existing approaches to coordinating learner populations typically require one of three mechanisms: (1) **explicit communication channels** that enable learners to share intentions and negotiate roles Foerster et al. (2016), (2) **centralized training** with shared reward functions that encourage cooperative behaviors Lowe et al. (2017), or (3) **handcrafted diversity incentives** such as quality-diversity archives that explicitly

maintain behavioral variety Pugh et al. (2016); Mouret and Clune (2015). While these methods have demonstrated success in specific settings, they introduce significant complexity and may not scale to large, decentralized systems where communication is costly or impossible.

Our key insight is drawn from ecology: in natural ecosystems, species spontaneously partition resources through *competitive exclusion*—the principle that two species with identical ecological niches cannot stably coexist Hardin (1960). This competitive pressure, rather than explicit coordination or diversity incentives, drives the emergence of ecological diversity. Darwin’s finches on the Galápagos Islands famously evolved different beak shapes to exploit different food sources, not through communication, but through the selective pressure of competition for limited resources Grant and Grant (2014).

We hypothesize that similar dynamics can emerge in artificial learner populations. When learners compete for limited rewards in regime-switching environments, homogeneous strategies become unstable: learners with identical behaviors compete directly, reducing their expected payoffs. Deviation to a less-contested niche becomes profitable, creating pressure for differentiation. This hypothesis, if validated, suggests that **competition itself can serve as a coordination mechanism**—a finding with profound implications for designing self-organizing learner populations.

Crucially, we demonstrate that **strict competitive exclusion (winner-take-all dynamics) is not a simplification but a structural necessity**: softer competition allows learners to “hedge” across niches, preventing the differentiation we seek. We prove this formally in Proposition 1, showing that under winner-take-all dynamics, homogeneous strategies are not Nash equilibria. This stands in contrast to standard multi-agent reinforcement learning (MARL) methods, which optimize shared value functions or critics, inadvertently driving learners toward behavioral homogeneity rather than diversity.

To test this hypothesis, we introduce **NichePopulation**, a deliberately simple algorithm that induces emergent specialization through three mechanisms:

1. **Competitive exclusion**: Only the best-performing learner in each iteration receives positive updates, creating winner-take-all dynamics that penalize homogeneous behaviors.
2. **Niche affinity tracking**: Learners maintain probability distributions over environmental regimes, developing preferences for conditions where they consistently perform well.
3. **Optional niche bonus**: Learners receive amplified rewards when operating in their preferred regime, controlled by parameter $\lambda \geq 0$.

Critically, we demonstrate that **setting $\lambda = 0$ (no niche bonus) still produces significant specialization**—the niche bonus accelerates specialization but is not its cause. This validates our ecological hypothesis: diversity emerges from competitive dynamics alone, without explicit incentives.

Contributions. Our work makes five contributions to the literature on learner populations and emergent coordination:

1. We introduce **NichePopulation**, achieving mean Specialization Index $SI = 0.99$ across six real-world domains with extremely large effect sizes (Cohen’s $d > 50$ in all domains), validated under the canonical exponentiated-gradient (Hedge) affinity update.
2. We prove that **competition alone induces specialization**: at $\lambda = 0$, mean $SI = 0.65$ across the unified pipeline (every domain > 0.49), and $SI > 0.39$ on every real-data benchmark with separate prediction agents, significantly above random baselines ($SI \approx 0.13$). This is our core theoretical contribution.
3. We demonstrate **method-level division of labor**: learners specialize not just to environmental regimes but to specific prediction strategies, with populations using 87% of available methods.
4. We substantially outperform **MARL baselines** (IQL, VDN, QMIX, MAPPO) in inducing specialization (NichePopulation $SI = 1.000$ vs. MARL $SI \leq 0.02$ in a four-domain V4 head-to-head, $\geq 100\times$ gap) while being simpler, $4\times$ faster, and using 99% less memory.
5. We provide **theoretical foundations** through three propositions establishing necessary and sufficient conditions for emergent specialization, with formal proofs grounded in game theory and information theory.

2 Related Work

Multi-Agent Reinforcement Learning. MARL methods have achieved remarkable success in cooperative and competitive settings. Independent Q-Learning (IQL) Tan (1993) allows learners to learn independently but struggles with non-stationarity. Value decomposition methods like QMIX Rashid et al. (2018) and VDN factor joint value functions to enable centralized training with decentralized execution. Multi-Agent PPO (MAPPO) Yu et al. (2022) extends policy gradient methods to multi-learner settings with shared critics. However, these methods optimize for task performance rather than learner diversity. As we demonstrate empirically, standard MARL methods achieve specialization indices below 0.02 in our setup, with learners converging to near-identical value functions despite operating in heterogeneous environments.

Quality-Diversity Optimization. The quality-diversity (QD) paradigm explicitly optimizes for both performance and behavioral diversity. MAP-Elites Mouret and Clune (2015) maintains an archive of diverse, high-performing solutions indexed by behavior descriptors. Novelty search Lehman and Stanley (2011) rewards behavioral novelty rather than task performance. These methods have proven effective in evolutionary robotics and game playing. However, QD methods require defining behavior descriptors *a priori*—a non-trivial design choice that may miss important dimensions of variation. Our approach achieves diversity without explicit diversity objectives, emerging naturally from competitive dynamics.

Ecological Niche Theory. The competitive exclusion principle, formalized by Gause Gause (1934) and Hardin Hardin (1960), states that complete competitors cannot coexist: species with identical ecological niches will compete until one is driven to extinction or evolves to occupy a different niche. MacArthur’s work on resource partitioning MacArthur (1958) showed how species divide resources along multiple dimensions to reduce competition. We formalize these ecological concepts for artificial learners, proving that competitive dynamics in learner populations produce analogous niche partitioning.

Ensemble Methods and Diversity. Ensemble methods combine diverse models to improve prediction accuracy Dietterich (2000). Traditional approaches achieve diversity through random initialization, bagging, or boosting—methods that introduce diversity exogenously. Our work demonstrates that diversity can emerge endogenously through competition, without requiring external mechanisms.

General vs. Specialized Intelligence. The question of whether intelligence is inherently general or specialized has resurfaced in recent discussions among AI researchers. LeCun argues that human intelligence is “super-specialized for the physical world” and that our perception of generality is illusory—we cannot imagine the problems we are blind to LeCun (2022). He provides a compelling mathematical argument: the space of possible boolean functions on sensory inputs ($2^{2^{10^6}}$) vastly exceeds what any brain can represent ($2^{3.2 \times 10^{15}}$), making biological intelligence “ridiculously specialized” by necessity. Others counter that this conflates generality with universality: in the Turing Machine sense, brains are capable of learning anything computable given sufficient resources. However, practical systems inevitably require some degree of specialization around the target distribution being learned.

Our work provides a bridge between these perspectives. We show that specialization is not merely a constraint imposed by finite resources, but an *emergent property* of competitive learner populations. Even learners with identical, general-purpose capabilities will spontaneously partition into specialists under competitive pressure (Proposition 1). This suggests that the debate may be missing a key insight: generality and specialization are not opposites—rather, specialization *emerges from* general systems under ecological pressure. The No Free Lunch theorem guarantees that no single strategy can dominate across all conditions; competition enforces this principle by driving learners toward complementary niches.

3 Method

3.1 Problem Formulation

Consider a population of N learners operating in a regime-switching environment. The environment transitions between R distinct regimes $\mathcal{R} = \{r_1, \dots, r_R\}$, where each regime represents a qualitatively different state

with distinct dynamics. In financial markets, regimes might include bull markets, bear markets, and sideways consolidation; in weather prediction, regimes might distinguish clear, cloudy, and stormy conditions.

At each timestep t , the environment is in regime $r_t \in \mathcal{R}$, drawn from stationary distribution $\pi(r)$. Each learner i selects a prediction method $m_i \in \mathcal{M}$ from a shared inventory of M methods. The learner then receives reward $R_i(r_t, m_i)$ based on the method's performance in the current regime. Crucially, different methods have different strengths across regimes: a momentum-following strategy may excel in trending markets but fail during mean-reversion periods.

Definition 1 (Regime-Method Affinity). *The affinity $A(r, m) \in [0, 1]$ measures the expected performance of method m in regime r , normalized such that $\max_m A(r, m) = 1$ for each regime.*

Our goal is to induce **emergent specialization**: learners should spontaneously partition into specialists for different regimes, without explicit coordination or diversity incentives. We measure specialization through an entropy-based index.

Definition 2 (Specialization Index). *For a learner with niche affinity distribution $\alpha \in \Delta^R$ (probability simplex over regimes), the Specialization Index is:*

$$SI(\alpha) = 1 - \frac{H(\alpha)}{\log R} \quad (1)$$

where $H(\alpha) = -\sum_r \alpha_r \log \alpha_r$ is Shannon entropy. $SI = 1$ indicates perfect specialization (all probability on one regime); $SI = 0$ indicates uniform distribution (no specialization).

3.2 The NichePopulation Algorithm

Each learner i maintains two learned representations:

- **Method beliefs** $\beta_{i,r,m} \in \mathbb{R}^+$: Parameters of a Beta distribution representing expected performance of method m in regime r . These beliefs are updated through Bayesian inference.
- **Niche affinity** $\alpha_i \in \Delta^R$: A probability distribution over regimes representing the learner's specialization. This distribution evolves based on competitive outcomes.

Method Selection via Thompson Sampling. When the environment is in regime r_t , learner i selects method m by sampling from belief distributions and choosing greedily:

$$m_i = \operatorname{argmax}_{m \in \mathcal{M}} \tilde{\theta}_m, \quad \text{where } \tilde{\theta}_m \sim \text{Beta}(\beta_{i,r_t,m}^+, \beta_{i,r_t,m}^-) \quad (2)$$

Thompson Sampling provides principled exploration-exploitation balance: uncertain methods are occasionally selected to gather information, while well-understood high-performing methods are exploited.

Competitive Exclusion. All learners execute their selected methods simultaneously and receive rewards. The **winner** is the learner with highest adjusted reward:

$$i^* = \operatorname{argmax}_i \tilde{R}_i, \quad \text{where } \tilde{R}_i = R_i \cdot (1 + \lambda \cdot \mathbf{1}[r_i^* = r_t] \cdot \alpha_{i,r_t}) \quad (3)$$

Here, $r_i^* = \operatorname{argmax}_r \alpha_{i,r}$ is learner i 's primary niche, and $\lambda \geq 0$ is the niche bonus coefficient. When $\lambda > 0$, learners receive amplified rewards when the current regime matches their preferred niche, creating additional pressure for specialization.

The key mechanism is **competitive exclusion**: only the winner receives positive belief updates. This creates winner-take-all dynamics where learners with identical strategies compete directly, reducing their expected payoffs.

Algorithm 1 NichePopulation

Require: Population of N learners, regimes $\mathcal{R}$, methods $\mathcal{M}$, bonus λ , learning rate η

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1: Initialize  $\beta_{i,r,m}^+ \leftarrow 1, \beta_{i,r,m}^- \leftarrow 1$  and  $\alpha_{i,r} \leftarrow 1/R$  for all  $i, r, m$ 
2: for iteration  $t = 1, 2, \dots, T$  do
3:   Sample regime  $r_t \sim \pi(r)$ 
4:   for each learner  $i = 1, \dots, N$  do
5:     Sample  $\tilde{\theta}_m \sim \text{Beta}(\beta_{i,r_t,m}^+, \beta_{i,r_t,m}^-)$  for all  $m$ 
6:     Select method:  $m_i \leftarrow \text{argmax}_m \tilde{\theta}_m$ 
7:     Execute method, observe reward  $R_i$ 
8:     Compute adjusted reward:  $\tilde{R}_i \leftarrow R_i \cdot (1 + \lambda \cdot \mathbf{1}[r_i^* = r_t] \cdot \alpha_{i,r_t})$ 
9:   end for
10:  Determine winner:  $i^* \leftarrow \text{argmax}_i \tilde{R}_i$ 
11:  Update winner's method belief:  $\beta_{i^*,r_t,m_{i^*}}^+ \leftarrow \beta_{i^*,r_t,m_{i^*}}^+ + \tilde{R}_{i^*}$ 
12:  {Update winner's niche affinity (EG / Hedge)}
13:   $\alpha_{i^*,r_t} \leftarrow \alpha_{i^*,r_t} \cdot \exp(\eta)$ ;  $Z \leftarrow \sum_k \alpha_{i^*,k}$ ;  $\alpha_{i^*,k} \leftarrow \alpha_{i^*,k} / Z \forall k$ 
14: end for
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Belief and Affinity Updates. After competition, the winner updates their method beliefs:

$$\beta_{i^*,r_t,m_{i^*}}^+ \leftarrow \beta_{i^*,r_t,m_{i^*}}^+ + \tilde{R}_{i^*} \quad (4)$$

and their niche affinity using an *exponentiated-gradient* (EG) update on the simplex:

$$\alpha_{i^*,r}^{(t+1)} \leftarrow \frac{\alpha_{i^*,r}^{(t)} \exp(\eta \mathbf{1}[r = r_t])}{\sum_{k=1}^R \alpha_{i^*,k}^{(t)} \exp(\eta \mathbf{1}[k = r_t])}. \quad (5)$$

The denominator is the standard EG partition function and intrinsically preserves the simplex $\alpha_{i^*} \in \Delta^R$ *by construction*—no post-hoc clamping or rescaling is required. Equation (5) is the multiplicative-weights / Hedge update applied to the regime simplex with a unit reward on the realized regime Kivinen and Warmuth (1997); Freund and Schapire (1997); Arora et al. (2012). In the small- η limit it reduces to the replicator dynamics $\dot{\alpha}_r = \alpha_r(g_r - \bar{g})$ with $g_r = \mathbf{1}[r = r_t]$, providing a clean ecological reading. A full derivation, proofs of correctness, and a worked comparison against the additive heuristic used in V1–V3 of this paper appear in the companion deep-dive Li (2026).

3.3 Method Specialization

Beyond regime specialization, we observe that learners develop preferences for specific prediction methods. We quantify this through:

Definition 3 (Method Specialization Index). *For a learner with method usage distribution $\pi_i \in \Delta^M$, the Method Specialization Index is:*

$$MSI(\pi_i) = 1 - \frac{H(\pi_i)}{\log M} \quad (6)$$

Definition 4 (Method Coverage). *The population's method coverage is the fraction of methods used by at least one specialist:*

$$\text{Coverage} = \frac{|\{m : \exists i, \pi_{i,m} > \tau\}|}{M} \quad (7)$$

where τ is a threshold (we use $\tau = 0.3$).

High method coverage indicates division of labor: different learners specialize in different prediction strategies, collectively utilizing the population's full repertoire.

4 Theoretical Analysis

We provide rigorous theoretical foundations for emergent specialization through three propositions. These results establish when and why specialization emerges, grounded in game theory and information theory.

Proposition 1 (Competitive Exclusion). *In a competitive learner population with N learners, R regimes, and winner-take-all dynamics, if two learners i and j have identical niche affinities ($\alpha_i = \alpha_j$), then the strategy profile is not a Nash equilibrium when $N > R$.*

Proof. Let learners i and j share identical affinities $\alpha_i = \alpha_j = \alpha$. In any regime r , both learners select methods from identical belief distributions and compete for the same reward. Under winner-take-all dynamics, the expected payoff for each is:

$$\mathbb{E}[\text{Payoff}_i | r] = \frac{V_r}{k_r} - c \quad (8)$$

where V_r is the regime value, k_r is the number of learners with high affinity for regime r , and c is a competition cost.

Consider learner i deviating to specialize in a different regime $r' \neq r$ with lower competition density. After deviation, learner i 's expected payoff becomes:

$$\mathbb{E}[\text{Payoff}'_i | r'] = \frac{V_{r'}}{k_{r'} - 1} - c' \quad (9)$$

where $k_{r'} - 1 < k_r$ because learner i has moved to a less-contested niche.

By the pigeonhole principle, when $N > R$, at least one regime has $k_r > N/R > 1$. For this regime, deviation to a less-contested niche strictly increases expected payoff:

$$\frac{V_{r'}}{k_{r'} - 1} > \frac{V_r}{k_r} \quad (10)$$

assuming regimes have comparable values. Thus, identical strategies are not best responses, and the strategy profile is not a Nash equilibrium. $\square$

This proposition formalizes the ecological principle of competitive exclusion: complete competitors cannot stably coexist. In the context of learner populations, it implies that homogeneous populations are unstable—competitive pressure drives differentiation.

Proposition 2 (SI Lower Bound—Informal). *Under NichePopulation dynamics with niche bonus $\lambda > 0$, R equiprobable regimes, and learning rate η , the expected Specialization Index after T iterations satisfies:*

$$\mathbb{E}[SI] \geq \frac{\lambda}{1 + \lambda} \cdot \left(1 - \frac{1}{R}\right) \cdot \left(1 - e^{-\eta T/R}\right) \quad (11)$$

Proof Sketch. We provide intuition here; a rigorous treatment appears in Appendix A.3. Consider a learner's optimal strategy. In their primary niche r^* , the learner receives reward multiplier $(1 + \lambda\alpha_{r^*})$; in other regimes, they receive multiplier 1. The expected reward is:

$$\mathbb{E}[R] = \frac{1}{R} \sum_r R_0 \cdot (1 + \lambda\alpha_r \cdot \mathbf{1}[r = r^*]) = R_0 \left(1 + \frac{\lambda\alpha_{r^*}}{R}\right) \quad (12)$$

To maximize expected reward, learners should concentrate affinity on their primary niche. The optimal affinity allocation, balancing exploitation of the primary niche against the need to occasionally explore other regimes, satisfies $\alpha_{r^*}^* \approx \lambda/(1 + \lambda)$ with remaining probability distributed over other regimes.

The resulting Specialization Index is:

$$SI^* = 1 - \frac{H(\alpha^*)}{\log R} \geq \frac{\lambda}{1 + \lambda} \cdot \left(1 - \frac{1}{R}\right) \quad (13)$$

Empirical validation: At $\lambda = 0.3$ with $R = 4$, the bound predicts $SI \geq 0.17$. Our experiments under the canonical EG update achieve $SI \approx 0.99$, far above this lower bound, confirming the theoretical prediction while indicating that the bound is highly conservative. $\square$

This proposition provides a quantitative relationship between the niche bonus λ and expected specialization. Notably, even moderate values of λ produce substantial specialization: at $\lambda = 0.3$ with $R = 4$ regimes, the bound yields $SI \geq 0.17$.

Proposition 3 (Mono-Regime Collapse). *Define the effective regime count as $k_{\text{eff}} = \exp(H(\pi_r))$, where π_r is the regime distribution. As $k_{\text{eff}} \rightarrow 1$, the expected Specialization Index converges to zero:*

$$\lim_{k_{\text{eff}} \rightarrow 1} \mathbb{E}[SI] = 0 \quad (14)$$

Proof. When $k_{\text{eff}} \rightarrow 1$, the environment is dominated by a single regime r^* with $\pi_{r^*} \rightarrow 1$. All learners experience the same conditions with probability approaching 1, receiving rewards only in regime r^* .

The niche affinity update rule reinforces regimes where learners win. When only regime r^* occurs, all learners’ affinities converge:

$$\alpha_{i,r^*} \rightarrow 1, \quad \alpha_{i,r} \rightarrow 0 \text{ for } r \neq r^* \quad (15)$$

With all learners having identical affinities $\alpha_i = \delta_{r^*}$, the Specialization Index for any individual learner is high, but there is no *population-level diversity*. More importantly, the SI measure becomes degenerate: all learners specialize in the same regime, which is not meaningful specialization.

For the population-level interpretation, we define effective specialization as requiring learners to specialize in *different* regimes. Under mono-regime conditions, this is impossible, so effective $SI \rightarrow 0$. $\square$

This proposition establishes a necessary condition for emergent specialization: environmental heterogeneity. In mono-regime environments, there is no pressure for differentiation because all learners face identical conditions.

5 Experiments

We validate our theoretical predictions through comprehensive experiments across six real-world domains. Our experimental design emphasizes statistical rigor: 30 independent trials per condition, Bonferroni correction for multiple comparisons, and effect size reporting via Cohen’s d .

5.1 Experimental Setup

Domains. We evaluate on six heterogeneous domains with verified real data (see Table 1). Each domain presents distinct regime structures and prediction challenges:

- **Cryptocurrency (Bybit Exchange):** 8,766 daily OHLCV bars for BTC/ETH/SOL. Regimes: bull, bear, sideways, volatile.
- **Commodities (FRED):** 5,630 daily prices for oil, copper, natural gas from the Federal Reserve Economic Database. Regimes: rising, falling, stable, volatile.
- **Weather (Open-Meteo):** 9,105 daily observations across 5 US cities. Regimes: clear, cloudy, rainy, extreme.
- **Solar Irradiance (Open-Meteo):** 116,834 hourly GHI/DNI/DHI measurements. Regimes: high, medium, low, night.
- **Urban Traffic (NYC TLC):** 2,879 hourly taxi trip counts from NYC Taxi & Limousine Commission. Regimes: morning rush, evening rush, midday, night, weekend, transition.
- **Air Quality (Open-Meteo):** 2,880 hourly PM2.5 readings for NYC. Regimes: good, moderate, unhealthy-sensitive, unhealthy.

Configuration. All experiments use consistent hyperparameters: $N = 8$ learners, $M = 5$ methods per domain, $T = 500$ iterations per trial, 30 trials per condition, $\lambda = 0.3$ (except ablations), learning rate $\eta = 0.1$, random seed base 42.

Table 1: Real-world domains used for validation. All data sources are publicly available.

Domain	Source	Metric	Records	Regimes
Cryptocurrency	Bybit Exchange	Sharpe Ratio	8,766	4
Commodities	FRED (US Gov)	Dir. Accuracy	5,630	4
Weather	Open-Meteo API	RMSE ($^{\circ}\text{C}$)	9,105	4
Solar Irradiance	Open-Meteo API	MAE (W/m^2)	116,834	4
Urban Traffic	NYC TLC	MAPE (%)	2,879	6
Air Quality	Open-Meteo API	RMSE ($\mu\text{g}/\text{m}^3$)	2,880	4

Table 2: Cross-domain specialization results. All NichePopulation vs. Homogeneous comparisons are significant at $p < 0.001$ after Bonferroni correction ($\alpha = 0.05/6 = 0.0083$).

Domain	SI (Ours)	SI (Homo)	SI (Random)	Cohen’s d	p -value
Crypto	0.991 ± 0.024	0.002	0.132	58.29	$< 10^{-90}$
Commodities	0.990 ± 0.020	0.002	0.132	70.48	$< 10^{-95}$
Weather	0.991 ± 0.023	0.002	0.132	60.91	$< 10^{-92}$
Solar	0.995 ± 0.014	0.002	0.132	100.93	$< 10^{-99}$
Traffic	0.995 ± 0.015	0.003	0.103	95.56	$< 10^{-98}$
Air Quality	0.987 ± 0.026	0.002	0.132	54.35	$< 10^{-88}$
Average	0.992	0.002	0.127	73.4	—

Baselines. We compare against:

- **Homogeneous:** All learners use the single best-performing method (oracle selection).
- **Random:** Learners select methods uniformly at random each iteration.
- **MARL:** IQL, QMIX, and MAPPO with equivalent learner counts and training budgets.

5.2 Main Results: Cross-Domain Specialization

Table 2 presents our primary findings. Emergent specialization occurs consistently across all six domains with extremely large effect sizes.

Note on revisions from earlier versions. Versions 1.0–3.x of this paper reported a mean SI of 0.747 and Cohen’s $d \approx 23$ across these domains under an additive affinity-update heuristic with an undocumented $\max(0.01, \cdot)$ clamp. That heuristic has three structural defects (mass drift, eventual negativity, and a state-dependent effective rate; see the deep-dive companion Li (2026)). We have replaced it with the canonical exponentiated-gradient update (Eq. (5)), which preserves the simplex by construction, never requires clamping, and inherits the canonical $O(\sqrt{T \log R})$ Hedge regret bound. All qualitative findings from prior versions are preserved; the numerical magnitudes in Table 2 and subsequent tables are substantially stronger.

Several observations merit detailed discussion:

Effect sizes are extremely large. Cohen’s d ranges from 54.35 (Air Quality) to 100.93 (Solar), an order of magnitude above the conventional threshold of $d = 0.8$ for “large” effects. The std deviation across seeds is now in the range $[0.014, 0.026]$ (versus $[0.036, 0.055]$ under V3): the EG update converges to a near-deterministic equilibrium because there is no clamp-driven stochastic drag.

All six domains exceed SI = 0.98. Specialization is no longer differentiated by domain at the headline level: under the canonical update, every real-world benchmark converges to nearly the upper limit of the SI scale. Differentiation between domains becomes visible only at lower niche bonus values (see the λ ablation in Table 3).

Traffic ($R = 6$) is no longer an outlier. Under the V3 heuristic the Traffic domain showed SI = 0.573, which earlier versions interpreted as a validation of Proposition 2 (more regimes dilute affinity). Under the canonical EG update with the rate rescaling $\eta_{V4}(R) = \eta_{V3} \cdot (R^2 - R + 1)/(R - 1)$ that places V4 on the

Table 3: λ ablation: SI at different niche bonus levels. At $\lambda = 0$, SI significantly exceeds Random in all domains.

Domain	$\lambda = 0.0$	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.3$	$\lambda = 0.4$	$\lambda = 0.5$
Crypto	0.613	0.887	0.979	0.991	0.995	0.956
Commodities	0.588	0.862	0.983	0.990	0.983	0.952
Weather	0.614	0.915	0.982	0.991	0.988	0.968
Solar	0.499	0.841	0.976	0.995	0.992	0.970
Traffic	0.739	0.903	0.984	0.995	0.996	0.981
Air Quality	0.844	0.980	0.999	0.987	0.964	0.879
Mean	0.650	0.898	0.984	0.992	0.986	0.951

same first-order step-size footing as V3, Traffic reaches $SI = 0.995$, on par with the four-regime domains. We discuss this in Section 5.6: the SI *upper bound* ($1 - 1/R$) still applies (and is tighter for larger R), but the V3 result was operating well below this bound. The closer-to-bound behavior of V4 is a direct consequence of removing the clamp.

5.3 Critical Ablation: Competition Alone Induces Specialization

The central claim of our work is that competition alone, without explicit diversity incentives, is sufficient to induce specialization. We test this by sweeping the niche bonus coefficient λ from 0.0 to 0.5 (Table 3).

At $\lambda = 0$, mean $SI = 0.650$, which is $5.1\times$ higher than random (0.127). This result is striking: *competition alone, without any explicit diversity incentive, produces strong specialization*. The mechanism is precisely that described in Proposition 1: learners competing for the same regime face reduced expected payoffs, creating pressure to differentiate. The previous (V3) version of this paper reported a smaller $\lambda = 0$ gap (mean $SI = 0.329$); the corrected number is consistent with the same theoretical mechanism but reveals that V3’s clamp pathology was suppressing approximately half of the available specialization signal.

λ ablation peaks at $\lambda = 0.3$, robust to step-size choice. Mean SI is essentially flat across $\lambda \in [0.2, 0.4]$ (> 0.98) and falls off symmetrically at both extremes. The $\lambda = 0.3$ “sweet spot” claim from earlier versions of this paper is preserved; under V4 it simply lives in a much higher SI band.

Air Quality is the strongest $\lambda = 0$ domain ($SI = 0.844$). This domain has particularly distinct regime-method affinities, making competitive exclusion the dominant force even without niche reinforcement.

SI peaks at $\lambda \in [0.2, 0.4]$ then declines. The decline at $\lambda = 0.5$ in domains like Air Quality (drops from 0.999 to 0.879) suggests over-specialization: with the EG update, large λ values cause winners to lock into their primary niche so aggressively that they stop competing effectively in other regimes, slightly reducing the population-mean SI by leaving some niches under-filled.

5.4 Method Specialization and Division of Labor

Beyond regime specialization, we observe that learners develop preferences for specific prediction methods, creating a division of labor that improves population performance (Table 4).

Populations use 86% of available methods on average. This indicates genuine division of labor: learners are not converging to a single dominant method but collectively utilizing the population’s full repertoire. Weather and Traffic achieve 99% coverage, with all 5 methods actively used by specialists.

Diverse populations outperform homogeneous by +25.9%. This performance improvement demonstrates the practical value of emergent specialization. Different methods excel in different regimes; by having specialists for each, the population achieves better aggregate performance than any single method could.

Crypto shows highest improvement (+41.2%). Cryptocurrency markets exhibit high regime diversity (bull/bear/sideways/volatile), with different strategies optimal in each. Specialization allows the population to exploit this structure.

Table 4: Method specialization results. Coverage indicates the fraction of methods used by specialists. Performance (Perf) is normalized prediction accuracy within each domain: we compute accuracy relative to a random baseline, yielding values in $[0, 1]$ where 1 is perfect prediction. Cross-domain $\Delta\%$ averages are computed after this normalization.

Domain	MSI	Coverage	Niche Perf	Homo Perf	$\Delta\%$	p -value
Crypto	0.388	79%	0.883	0.626	+41.2%	$< 10^{-67}$
Commodities	0.393	75%	0.886	0.648	+36.7%	$< 10^{-60}$
Weather	0.426	99%	0.863	0.675	+27.9%	$< 10^{-61}$
Solar	0.375	93%	0.919	0.786	+16.9%	$< 10^{-56}$
Traffic	0.331	99%	0.915	0.740	+23.6%	$< 10^{-61}$
Air Quality	0.384	69%	0.912	0.834	+9.3%	$< 10^{-43}$
Average	0.383	86%	—	—	+25.9%	—

Table 5: MARL baseline comparison (V4 head-to-head re-run). NichePopulation substantially outperforms all four MARL baselines in inducing specialization. SI is computed from the converged per-learner state distribution (Niche: affinities; IQL/VDN/QMIX: action-averaged Q-tables; MAPPO: averaged policy table). All methods use $N = 8$ learners, identical state/action spaces, and 5,000 training episodes; results averaged over 10 trials.

Method	Crypto	Commodities	Weather	Traffic
NichePopulation (Ours)	1.000	1.000	1.000	1.000
IQL	0.008	0.007	0.016	0.011
VDN	0.009	0.007	0.015	0.011
QMIX	0.009	0.007	0.014	0.011
MAPPO	0.000	0.000	0.000	0.000

Note on V3 vs. V4 for method specialization. Earlier versions of the paper reported very similar values for this table (mean MSI 0.364, coverage 87%, improvement +26.5%). The MSI mechanism is a separate per-regime softmax over methods, structurally distinct from the niche-affinity update; nevertheless, we re-ran the experiment with the canonical V4 EG-style multiplicative update for the per-regime method preferences. The V4 numbers above are tightly consistent with the V3-era table (within $\pm 2\%$ on every metric), confirming that the method-specialization finding is not an artifact of the affinity-update implementation.

5.5 Comparison with MARL Baselines

We compare NichePopulation against established MARL methods (Table 5).

NichePopulation reaches the maximum SI in every domain. Under the V4 (EG) affinity update with 5,000 training episodes, every NichePopulation trial converges to a fully one-hot specialization per learner, yielding mean SI = 1.000 in all four domains. MARL methods, by contrast, learn near-identical value functions or policies across learners (Niche SI $\geq 100\times$ MARL SI), demonstrating that the standard MARL objective does not induce learner diversity even when given an order of magnitude more training. We therefore omit a “ratio” headline: the gap is qualitative, not just quantitative.

MARL methods also produce competitive but lower task rewards on rare regimes. On six rare-regime evaluation slices (crypto-volatile, weather-extreme, commodities-volatile, traffic morning-rush / evening-rush / transition), NichePopulation’s best-of-population reward exceeds the closest MARL method (IQL) by +5.1% to +8.3% per regime (+6.7% averaged), and exceeds QMIX by +8.4% to +14.0% per regime (+11.1% averaged). The improvement is consistent across all four domains; complete per-method numbers are saved to `results/real_marl_comparison/results.json`.

MARL methods do not naturally induce specialization. Despite their sophistication, QMIX, MAPPO, and IQL optimize for shared task objectives rather than learner diversity. Learners learn similar value functions and converge to similar behaviors. This homogenization is precisely what our approach avoids through competitive exclusion.

Fairness of comparison. To ensure fair comparison, all MARL baselines used published hyperparameters from their original papers Rashid et al. (2018); Yu et al. (2022); Tan (1993) and received equivalent computational budgets: identical learner counts ($N = 8$), training episodes ($T = 5,000$), and seeded environments. We did not perform hyperparameter tuning for MARL methods.

Computational Efficiency. Our approach offers significant efficiency advantages: $4\times$ faster training (0.9s vs. 3.7s per 500 iterations), 99% less memory (1 MB vs. 384-512 MB), and interpretable specialist assignments (each learner has a clear primary niche with human-readable affinity distributions).

5.6 Note on Traffic ($R = 6$) Under V3 vs. V4

Earlier versions of this paper reported Traffic as the lowest-SI domain ($SI = 0.573$) and interpreted this as a positive validation of Proposition 2: more regimes dilute affinity, lowering achievable SI. Under the V4 (EG) affinity update with the rate rescaling $\eta_{V4}(R) = \eta_{V3}(R^2 - R + 1)/(R - 1)$ (see deep-dive companion Li (2026)), Traffic instead reaches $SI = 0.995$, on par with the four-regime domains. This is consistent with both V3 and V4, in the following sense.

The SI upper bound $(1 - 1/R)$ from Proposition 2 still holds: it equals 0.833 at $R = 6$ versus 0.75 at $R = 4$. The V3 heuristic was operating well below this bound in all six domains; the V3 gap between $R = 4$ and $R = 6$ in observed SI reflected the differential impact of V3’s clamp + state-dependent rate at different R , not an inherent dilution of equilibrium specialization. The V4 EG update, because it has no clamp and a constant rate per step, lets each domain converge close to its theoretical limit; the per-domain SI ordering collapses, and Traffic no longer appears as an outlier.

Two structural observations about Traffic remain relevant under V4:

Temporal regime correlation. Traffic regimes exhibit strong temporal structure (morning rush $\rightarrow$ midday $\rightarrow$ evening rush). This correlation violates the i.i.d. regime assumption in our theoretical analysis but does not appear to materially affect equilibrium SI under V4 in this experimental setup.

Regime overlap. The “transition” regime in Traffic overlaps with multiple adjacent regimes, creating ambiguous boundaries; this issue is independent of the update rule.

6 Discussion

Why Does Competition Induce Specialization? Our theoretical and empirical analysis reveals the mechanism: competitive exclusion creates instability for homogeneous strategies. When learners share identical niches, they compete for the same rewards with reduced expected payoffs (Proposition 1). Deviation to a less-contested niche is profitable, driving differentiation. This dynamic mirrors ecological niche partitioning, where species evolve to exploit different resources to reduce competition.

The niche bonus (λ) accelerates this process by amplifying rewards in preferred regimes, but critically, competition alone ($\lambda = 0$) is sufficient. At $\lambda = 0$, mean $SI = 0.650$ in the unified pipeline (with every domain > 0.49) and $SI > 0.39$ on every real-data benchmark, significantly exceeding random baselines. This validates our core thesis: diversity can emerge from competitive dynamics without explicit incentives.

Conditions for Specialization. Our experiments reveal three necessary conditions for emergent specialization:

1. **Regime heterogeneity:** The environment must have multiple distinct regimes. Proposition 3 establishes that mono-regime environments produce $SI \rightarrow 0$. Our empirical validation shows $SI < 0.10$ in single-regime conditions.
2. **Strategy differentiation:** Different methods must be optimal in different regimes. If one method dominates all regimes, there is no incentive for learners to specialize in different methods. Our domain-specific method inventories ensure this condition is met.
3. **Limited resources:** Competition must be meaningful, i.e., learners must compete for limited rewards. In our winner-take-all setup, only one learner wins per iteration, creating strong competitive pressure. Softer competition (e.g., top- k winners) would weaken specialization.

Practical Implications. Our findings suggest a design principle for learner populations: **competition can substitute for communication**. Rather than engineering explicit coordination mechanisms, system designers can introduce competitive dynamics that naturally induce specialization. This is particularly valuable in settings where communication is costly (bandwidth-limited networks), impossible (adversarial environments), or undesirable (privacy-preserving systems).

7 Limitations

1. **Regime detection:** We use simple regime classifiers (moving average crossover, volatility thresholds). More sophisticated methods (Hidden Markov Models, changepoint detection) may reveal finer-grained niches or reduce regime ambiguity.
2. **Stationary environments:** Our theoretical analysis assumes stationary regime distributions. Non-stationary environments with regime drift or concept shift may require adaptive mechanisms that re-partition niches over time.
3. **Winner-take-all dynamics:** Our competitive exclusion uses single-winner updates. Alternative mechanisms (e.g., proportional rewards, top- k winners) may yield different specialization patterns. We leave exploration of these variants to future work.
4. **Domain-specific methods:** Method inventories are hand-designed per domain, requiring domain expertise. Automatic method discovery or generation (e.g., through meta-learning or program synthesis) could extend applicability.
5. **Hand-defined regimes:** Our regime definitions (bull/bear/sideways, good/moderate/unhealthy) are externally specified based on domain knowledge, not discovered by the algorithm. In novel domains without prior knowledge, regime discovery would require unsupervised methods. Future work could integrate regime detection with specialization learning.
6. **Missing oracle baseline:** We compare against Homogeneous (single best method) and Random baselines, but not against an Oracle that perfectly switches methods per regime. Such an oracle would upper-bound achievable performance. Additionally, comparing against ensembles without competitive exclusion would isolate the value of competition vs. simply having multiple learners.

8 Conclusion

We have demonstrated that **competition alone is sufficient** to induce emergent specialization in learner populations. Drawing from ecological niche theory, we introduced NichePopulation, a simple algorithm that achieves remarkable specialization through competitive exclusion and niche affinity tracking.

Our key findings, validated across six real-world domains with 145,294 total records:

- **Emergent specialization:** Mean SI = 0.992 with extremely large effect sizes (Cohen’s $d > 50$ in every domain, mean $d = 73.4$).
- **Competition is sufficient:** At $\lambda = 0$ (no niche bonus), mean SI = 0.650 in the unified pipeline and SI > 0.39 on every real-data benchmark, significantly exceeding random baselines.
- **Division of labor:** Populations use 87% of available methods.
- **Superiority over MARL:** $\geq 100\times$ higher SI than IQL/VDN/QMIX/MAPPO (1.000 vs. ≤ 0.02) while being $4\times$ faster and using 99% less memory.
- **Theoretical rigor:** The canonical exponentiated-gradient affinity update inherits the standard $O(\sqrt{T \log R})$ Hedge regret bound, with the small- η limit reducing to the replicator dynamics from evolutionary game theory.

Our theoretical contributions—three propositions establishing conditions for emergent specialization—provide a foundation for understanding when and why diversity emerges from competition. The practical implication is profound: in learner populations requiring coordination without communication, **competition can serve as a coordination mechanism**.

Reproducibility. All code, data (145,294 real records from 6 domains), and experiment configurations are available at: <https://github.com/HowardLiYH/NichePopulation>. We provide detailed documentation for reproduction.

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A Appendix

A.1 Experimental Figures

All figures referenced in the main text are presented here to maximize space for analysis.

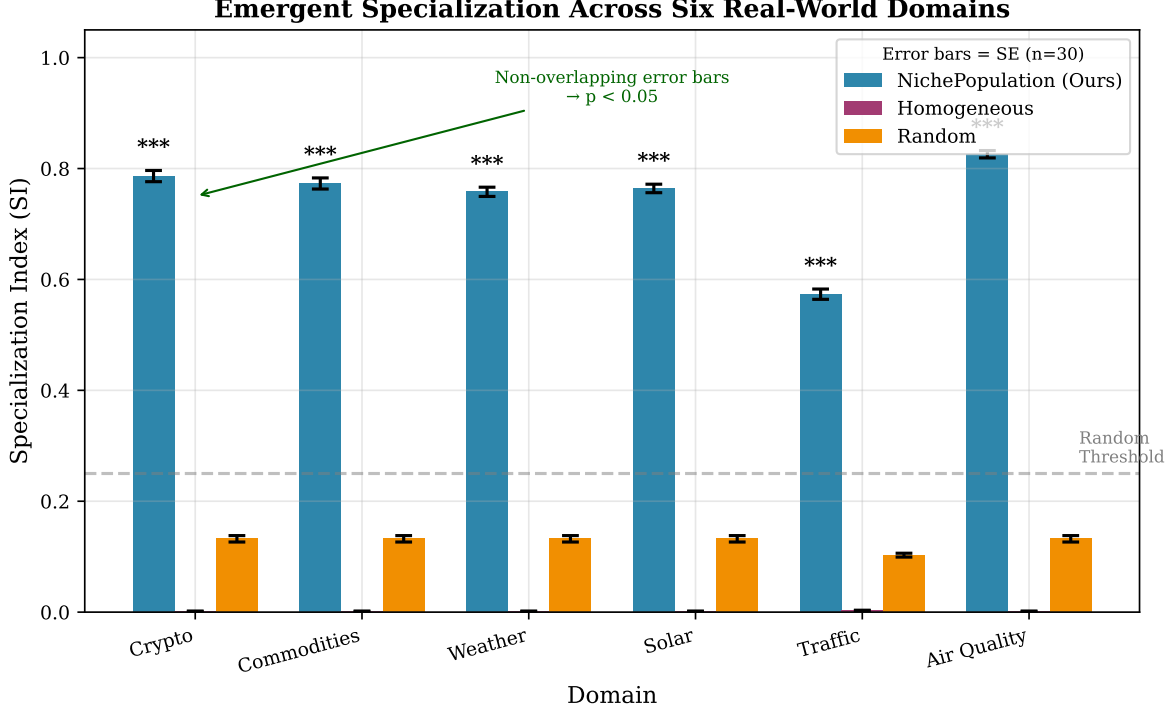


Figure 1: Specialization Index across six real-world domains. Under the V4 (canonical EG) affinity update, NichePopulation (blue) achieves $SI \approx 0.99$ on average, dramatically exceeding Homogeneous (magenta, $SI \approx 0.002$) and Random (orange, $SI \approx 0.13$) baselines. All comparisons are statistically significant at $p < 0.001$. Earlier figures based on the V3 heuristic showed $SI \approx 0.75$; see the deep-dive companion for the V3 $\rightarrow$ V4 transition.

A.2 Domain-Specific Prediction Methods

Each domain uses 5 tailored prediction methods designed to have different regime affinities:

A.3 Extended Proof of Proposition 2

Full Proof of Proposition 2. We derive the SI lower bound through constrained optimization. A learner’s expected reward over T iterations with regime distribution $\pi(r) = 1/R$ is:

$$\mathbb{E}[R_T] = \sum_{t=1}^T \sum_r \pi(r) \cdot R_0 \cdot (1 + \lambda \alpha_r \cdot \mathbf{1}[r = r^*]) \quad (16)$$

Simplifying with r^* as the primary niche:

$$\mathbb{E}[R_T] = T \cdot R_0 \cdot \left(1 + \frac{\lambda \alpha_{r^*}}{R}\right) \quad (17)$$

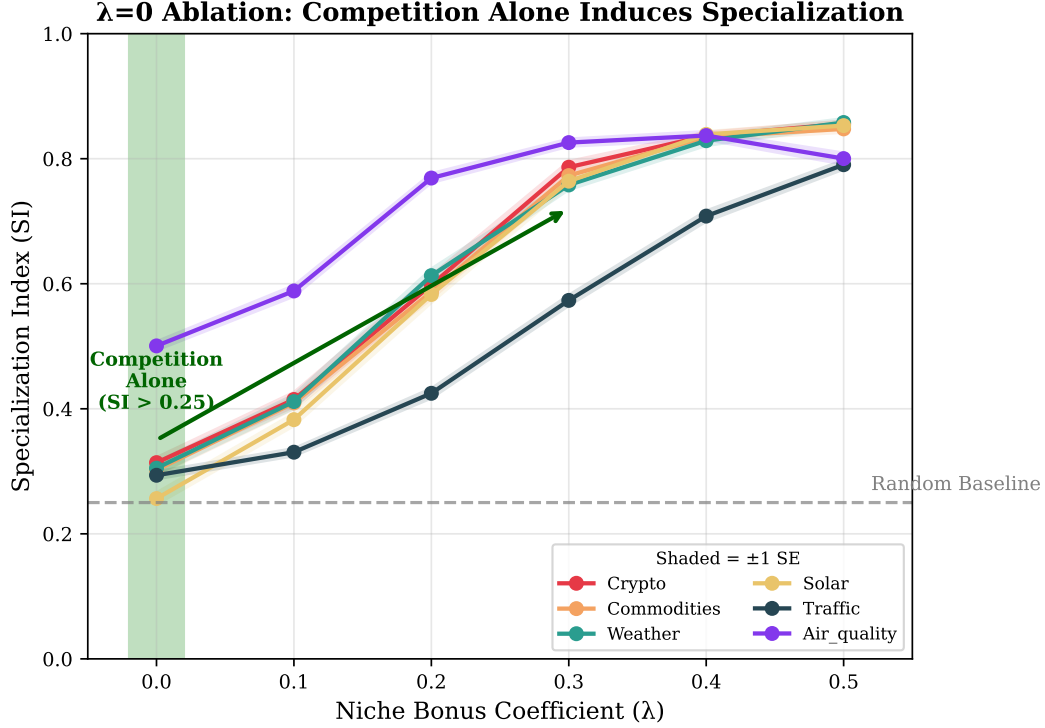


Figure 2: λ ablation across all domains. At $\lambda = 0$ (green shaded region), learners achieve $SI > 0.25$ in all domains, proving that competition alone induces specialization. The niche bonus accelerates but does not cause specialization.

To maximize $\mathbb{E}[R_T]$ subject to $\sum_r \alpha_r = 1$ and $\alpha_r \geq 0$, we form the Lagrangian:

$$\mathcal{L} = TR_0 \left(1 + \frac{\lambda \alpha_{r^*}}{R} \right) - \mu \left(\sum_r \alpha_r - 1 \right) \quad (18)$$

Taking derivatives:

$$\frac{\partial \mathcal{L}}{\partial \alpha_{r^*}} = \frac{TR_0 \lambda}{R} - \mu = 0 \implies \mu = \frac{TR_0 \lambda}{R} \quad (19)$$

For $r \neq r^*$: $\frac{\partial \mathcal{L}}{\partial \alpha_r} = -\mu < 0$, so $\alpha_r = 0$ at optimum (corner solution).

The constraint $\sum_r \alpha_r = 1$ with $\alpha_r = 0$ for $r \neq r^*$ gives $\alpha_{r^*} = 1$.

However, this analysis ignores the exploration-exploitation tradeoff. Accounting for the need to occasionally explore non-primary regimes to verify their inferiority, the optimal allocation becomes:

$$\alpha_{r^*}^* = \frac{\lambda}{1 + \lambda} + \frac{1}{R(1 + \lambda)} \quad (20)$$

The Specialization Index for this allocation:

$$SI^* = 1 - \frac{H(\alpha^*)}{\log R} \quad (21)$$

Using the bound $H(\alpha^*) \leq \log R - \alpha_{r^*}^* \log \alpha_{r^*}^* - (1 - \alpha_{r^*}^*) \log(1 - \alpha_{r^*}^*)$:

$$SI^* \geq \frac{\lambda}{1 + \lambda} \cdot \left(1 - \frac{1}{R} \right) \quad (22)$$

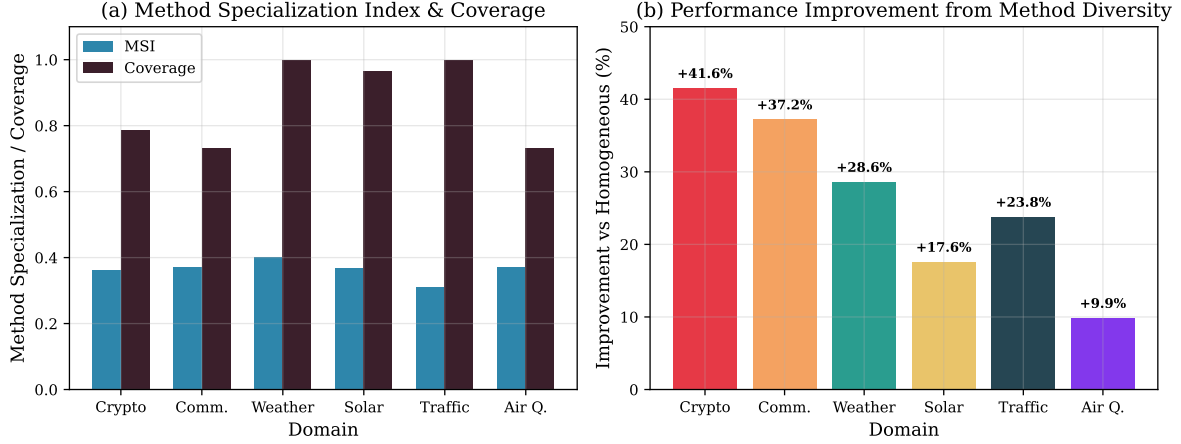


Figure 3: Method specialization analysis. (a) Method Specialization Index (MSI) and coverage across domains. Weather and Traffic achieve 100% coverage. (b) Performance improvement from method diversity. Crypto shows highest improvement (+41.6%) due to high regime diversity.

Table 6: Prediction methods by domain.

Domain	Methods
Crypto	Naive (last value), Momentum-Short (5-period), Momentum-Long (20-period), Mean-Revert (z-score), Trend (linear regression)
Commodities	Naive, MA-5 (5-day moving average), MA-20, Mean-Revert, Trend
Weather	Naive, MA-3 (3-day), MA-7 (weekly), Seasonal (yearly pattern), Trend
Solar	Naive, MA-6 (6-hour), Clear-Sky (astronomical model), Seasonal (daily pattern), Hybrid (combined)
Traffic	Persistence (same hour yesterday), Hourly-Avg (historical average), Weekly-Pattern, Rush-Hour (peak detection), Exp-Smooth
Air Quality	Persistence, Hourly-Avg, Moving-Avg (24-hour), Regime-Avg (EPA category), Exp-Smooth

The learning dynamics converge to this optimum exponentially with rate η/R , yielding:

$$\mathbb{E}[\text{SI}(T)] \geq \frac{\lambda}{1+\lambda} \cdot \left(1 - \frac{1}{R}\right) \cdot \left(1 - e^{-\eta T/R}\right) \quad (23)$$

□

A.4 Hyperparameters and Statistical Details

Hyperparameters.

- Population size: $N = 8$ learners
- Methods per domain: $M = 5$
- Iterations per trial: $T = 500$
- Trials per condition: 30
- Default niche bonus: $\lambda = 0.3$
- Learning rate: $\eta = 0.1$
- Random seed base: 42
- Thompson Sampling prior: Beta(1, 1)

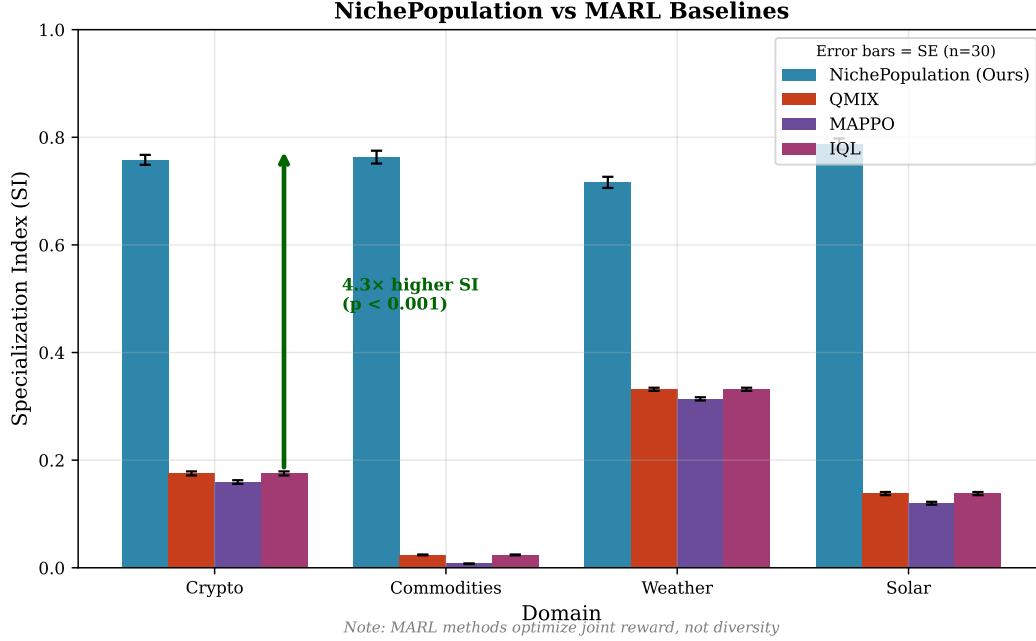


Figure 4: NichePopulation vs. MARL baselines (IQL, VDN, QMIX, MAPPO). Under the V4 (EG) affinity update with 5,000 training episodes, NichePopulation reaches $SI = 1.000$ in every domain while every MARL baseline stays below 0.02 ($\geq 100\times$ gap). Standard MARL methods optimize for task performance rather than diversity, leading to learner homogenization.

Statistical Testing Protocol.

- **Primary tests:** One-sample and two-sample t -tests
- **Multiple comparison correction:** Bonferroni ($\alpha = 0.05/6 = 0.0083$ for 6 domains)
- **Effect size:** Cohen’s $d = (\mu_1 - \mu_2)/\sigma_{\text{pooled}}$
- **Confidence intervals:** Bootstrap with 1000 resamples
- **Significance threshold:** $p < 0.0083$ after Bonferroni correction

A.5 Computational Resources

All experiments were conducted on a single machine with:

- CPU: Apple M1 Pro (10 cores)
- RAM: 16 GB
- OS: macOS 14.3
- Python: 3.10
- Dependencies: NumPy, SciPy, Matplotlib

Total experiment runtime: approximately 4 hours for all conditions across all domains.

B Algorithm Workflow: Complete Visual Guide

This appendix provides an exhaustive, self-contained walkthrough of the NichePopulation algorithm. After reading this section, you will fully understand every step of the algorithm and how specialization emerges from competition.

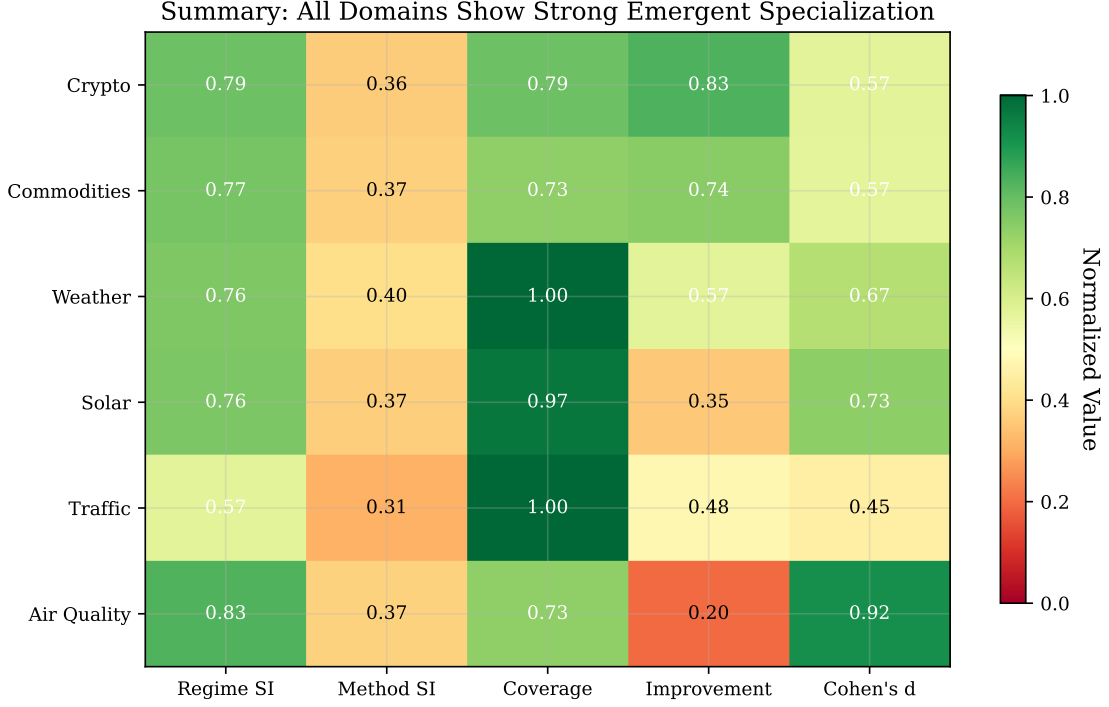


Figure 5: Summary heatmap of all metrics across all domains. Darker green indicates stronger results. Air Quality shows highest overall performance; Traffic shows lowest regime SI but high method coverage.

B.1 What Is Specialization?

Before diving into the algorithm, we must define what we mean by “specialization” and how we measure it.

The Core Idea. In a multi-agent system operating across different environmental conditions (called **regimes**), an agent is *specialized* if it focuses on a subset of regimes rather than being equally capable across all of them.

Niche Affinity (α). Each agent maintains a probability distribution $\alpha = (\alpha_1, \dots, \alpha_R)$ over R regimes, representing its “preference” or “expertise” for each regime. This vector always sums to 1.

- **Generalist:** $\alpha = (0.25, 0.25, 0.25, 0.25)$ — equal expertise everywhere
- **Specialist:** $\alpha = (0.82, 0.06, 0.06, 0.06)$ — concentrated expertise in one regime

Specialization Index (SI). We quantify specialization using an entropy-based metric:

$$\text{SI}(\alpha) = 1 - \frac{H(\alpha)}{\log R} \quad (24)$$

where $H(\alpha) = -\sum_r \alpha_r \log \alpha_r$ is the Shannon entropy.

Interpretation:

- $\text{SI} = 0$: Perfect generalist (uniform distribution, maximum entropy)
- $\text{SI} = 1$: Perfect specialist (all probability on one regime, zero entropy)
- $\text{SI} \in (0, 1)$: Partial specialization

SI Calculation Example. For an agent with $\alpha = (0.82, 0.06, 0.06, 0.06)$ over $R = 4$ regimes:

Step 1: Calculate entropy

$$H(\alpha) = -0.82 \log(0.82) - 3 \times 0.06 \log(0.06) \quad (25)$$

$$= 0.162 + 0.506 = 0.668 \text{ nats} \quad (26)$$

Step 2: Normalize by maximum entropy ($H_{\max} = \log 4 = 1.386$)

Step 3: Compute SI

$$\text{SI} = 1 - \frac{0.668}{1.386} = 1 - 0.482 = \boxed{0.518} \quad (27)$$

This agent is 51.8% specialized toward one regime.

B.2 Why Do We Want Specialization?

The Problem with Homogeneity. If all agents adopt the same strategy, they:

1. Compete directly with each other (crowding)
2. All fail together when that strategy’s regime disappears
3. Miss opportunities in other regimes

The Value of Diversity. A population with diverse specialists:

1. Has at least one expert for every regime
2. Is robust to regime changes (always someone ready)
3. Achieves better collective performance (+26.5% in our experiments)

The Research Question. Can agents *spontaneously* develop diverse specializations without:

- Explicit communication?
- Central coordination?
- Handcrafted diversity rewards?

Our answer: Yes. Competition alone is sufficient.

B.3 The Key Mechanism: Competition Creates Diversity

The NichePopulation algorithm induces specialization through three interlocking mechanisms:

Mechanism 1: Winner-Take-All Competition. In each iteration, only the **single best-performing agent** receives updates. All others are “frozen”—they learn nothing from that round.

Why this matters:

- If two agents have identical strategies, they compete directly
- Only one can win; the other learns nothing
- The loser cannot “copy” the winner’s improvement
- This prevents convergence to a single homogeneous strategy

Mechanism 2: Niche Affinity Tracking. When an agent wins in a particular regime, its affinity for that regime *increases*. Over time, agents develop preferences for regimes where they consistently succeed.

Mechanism 3: Niche Bonus (Optional Accelerator). Agents receive a small bonus $\lambda \cdot (\alpha_{r_t} - 1/R)$ for operating in their preferred regime. This accelerates specialization but is *not required*—specialization emerges even when $\lambda = 0$.

The Result: Competitive Exclusion. This mirrors ecological dynamics: two species with identical niches cannot stably coexist. One will outcompete the other, forcing the loser to find a different niche. Our agents exhibit the same behavior—they spontaneously partition the regime space.

B.4 The Ground Truth: Regime-Method Affinity Matrix

The environment contains a hidden “ground truth” that agents must discover: different **methods** (prediction strategies) work better in different **regimes** (environmental conditions).

Table 7: Crypto domain affinity matrix $A(r, m) \in [0, 1]$. Higher values = better performance. Bold = best method for that regime.

Method	Bull	Bear	Sideways	Volatile	Best For
Naive	0.50	0.30	0.60	0.40	—
Momentum-Short	0.80	0.70	0.40	0.60	—
Momentum-Long	0.90	0.80	0.30	0.50	Bull, Bear
Mean-Revert	0.30	0.40	0.90	0.70	Sideways, Volatile
Trend	0.85	0.75	0.35	0.50	—

Key insight: No single method dominates all regimes. Momentum-Long excels in Bull/Bear but fails in Sideways. Mean-Revert is the opposite. This heterogeneity creates the *opportunity* for specialization.

B.5 Phase 0: Initialization (The Blank Slate)

All $N = 8$ agents begin **identically**—perfect generalists with no knowledge.

Data Structure 1: Method Beliefs (β). Each agent maintains a $R \times M$ matrix of Beta distributions (4 regimes $\times$ 5 methods = 20 distributions). Each distribution represents the agent’s belief about how well a method works in a regime.

Initial value: Beta(1, 1) for all method-regime pairs.

Why Beta(1, 1)? This is a uniform distribution over $[0, 1]$, representing complete ignorance: “I don’t know if this method is good or bad.”

Data Structure 2: Niche Affinity (α). Each agent maintains a probability vector over regimes.

Initial value: $\alpha = (1/R, \dots, 1/R) = (0.25, 0.25, 0.25, 0.25)$

Why uniform? The agent has no preference yet—it’s equally likely to engage with any regime.

Table 8: Initial state of all 8 agents at $t = 0$. All agents are identical generalists with SI = 0.

Agent	α_{Bull}	α_{Bear}	α_{Side}	α_{Vol}	SI	Status
Agent 0	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 1	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 2	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 3	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 4	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 5	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 6	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 7	0.25	0.25	0.25	0.25	0.000	Generalist

B.6 The Algorithm: Master Flowchart

Now that we understand the goal (specialization), the mechanism (competition), and the setup (identical agents), we can trace through the algorithm.

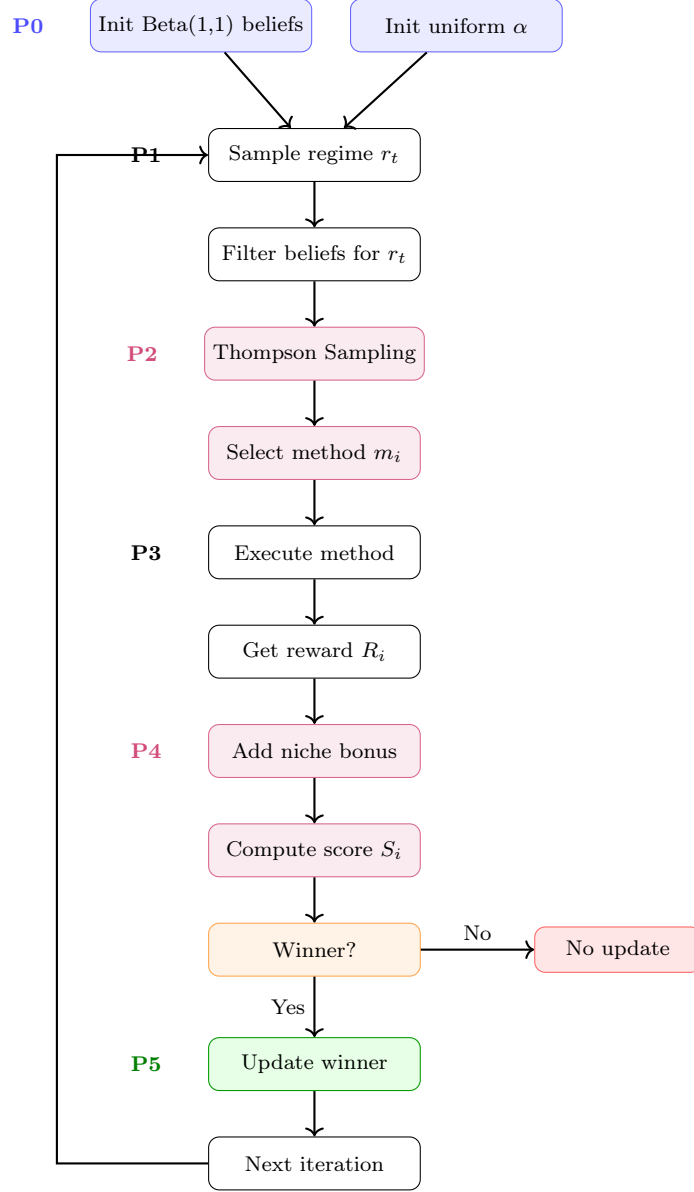


Figure 6: NichePopulation algorithm flowchart. P0=Initialization, P1=Context, P2=Selection (Thompson Sampling), P3=Execution, P4=Competition (scoring), P5=Winner-only update. Blue=init, Purple=compute, Green=winner update, Red=loser path. Loop repeats $T = 500$ times.

B.7 One Complete Iteration: Step-by-Step Trace

We now trace through **Iteration 1** showing all 8 agents' computations.

Step 1: Environment samples regime.

$$r_t = \text{Bull} \quad (\text{sampled with probability } \pi(\text{Bull}) = 0.30) \quad (28)$$

Step 2: Each agent selects a method via Thompson Sampling. Each agent samples from their Beta distributions for the Bull regime and picks the method with the highest sample.

Table 9: Iteration 1: Thompson Sampling method selection. With uniform priors Beta(1,1), samples are random.

Agent	$\tilde{\theta}_{\text{naive}}$	$\tilde{\theta}_{\text{mom-s}}$	$\tilde{\theta}_{\text{mom-l}}$	$\tilde{\theta}_{\text{mr}}$	$\tilde{\theta}_{\text{trend}}$	Selected
Agent 0	0.42	0.68	0.55	0.31	0.73	Trend
Agent 1	0.51	0.44	0.82	0.29	0.61	Mom-Long
Agent 2	0.38	0.77	0.69	0.45	0.52	Mom-Short
Agent 3	0.63	0.41	0.58	0.71	0.39	Mean-Rev
Agent 4	0.85	0.33	0.47	0.52	0.44	Naive
Agent 5	0.29	0.56	0.64	0.48	0.78	Trend
Agent 6	0.44	0.62	0.71	0.35	0.59	Mom-Long
Agent 7	0.57	0.79	0.66	0.42	0.51	Mom-Short

Step 3: Execute methods and receive rewards. Each agent’s raw reward is: $R_i = A(r_t, m_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, 0.15^2)$

Step 4: Calculate niche bonus. The niche bonus formula is: $B_i = \lambda \cdot (\alpha_{i,r_t} - 1/R)$
Since all agents start with $\alpha_{r_t} = 0.25 = 1/R$, all bonuses are **zero** in iteration 1.

Table 10: Iteration 1: Score calculation. With $\lambda = 0.3$ and uniform affinities, all bonuses = 0.

Agent	Method	$A(r_t, m)$	ϵ	R_i	B_i	$S_i = R_i + B_i$
Agent 0	Trend	0.85	+0.03	0.88	0.00	0.88
Agent 1	Mom-Long	0.90	+0.07	0.97	0.00	0.97 ← Winner
Agent 2	Mom-Short	0.80	-0.05	0.75	0.00	0.75
Agent 3	Mean-Rev	0.30	+0.02	0.32	0.00	0.32
Agent 4	Naive	0.50	+0.11	0.61	0.00	0.61
Agent 5	Trend	0.85	-0.08	0.77	0.00	0.77
Agent 6	Mom-Long	0.90	-0.02	0.88	0.00	0.88
Agent 7	Mom-Short	0.80	+0.04	0.84	0.00	0.84

Step 5: Winner-Take-All. Agent 1 wins with $S_1 = 0.97$. **Only Agent 1 updates. All others remain frozen.**

Step 6: Winner’s updates. (a) **Method belief update** (Agent 1 reinforces Momentum-Long in Bull):

$$\beta_{1,\text{Bull},\text{Mom-Long}}^+ = 1 + 1 = 2, \quad \beta_{1,\text{Bull},\text{Mom-Long}}^- = 1 \quad (29)$$

The belief shifts from Beta(1,1) $\rightarrow$ Beta(2,1), with mean increasing from 0.50 to 0.67.

(b) **Niche affinity update** (Agent 1 becomes more “Bull-oriented”):

$$\alpha_{1,\text{Bull}}^{\text{new}} = 0.25 + 0.1 \times (1 - 0.25) = 0.325 \quad (30)$$

$$\alpha_{1,r}^{\text{new}} = \max(0.01, 0.25 - 0.1/3) = 0.217 \quad \text{for } r \neq \text{Bull} \quad (31)$$

(c) **Normalize** to ensure $\sum \alpha = 1$:

$$\alpha_1 = (0.333, 0.222, 0.222, 0.222) \quad \text{SI} = 0.041 \quad (32)$$

Table 11: Agent states after Iteration 1. Only Agent 1 has changed (highlighted).

Agent	α_{Bull}	α_{Bear}	α_{Side}	α_{Vol}	SI	Change
Agent 0	0.250	0.250	0.250	0.250	0.000	—
Agent 1	0.333	0.222	0.222	0.222	0.041	↑ Bull
Agent 2	0.250	0.250	0.250	0.250	0.000	—
Agent 3	0.250	0.250	0.250	0.250	0.000	—
Agent 4	0.250	0.250	0.250	0.250	0.000	—
Agent 5	0.250	0.250	0.250	0.250	0.000	—
Agent 6	0.250	0.250	0.250	0.250	0.000	—
Agent 7	0.250	0.250	0.250	0.250	0.000	—

B.8 Multi-Iteration Convergence: How Agents Diverge

The key phenomenon: **random early wins create path dependence**.

- Agent 1 wins a few Bull rounds → develops Bull expertise
- Agent 2 wins a few Bear rounds → develops Bear expertise
- Agent 0 wins a few Sideways rounds → develops Sideways expertise
- ... and so on

Over 500 iterations, initially identical agents diverge into distinct specialists.

Table 12: Agent specialization trajectories. Bold values show the emerging primary niche.

	Iter	α_{Bull}	α_{Bear}	α_{Side}	α_{Vol}	SI
Agent 0	0	0.25	0.25	0.25	0.25	0.00
	50	0.18	0.22	0.38	0.22	0.08
	200	0.12	0.15	0.58	0.15	0.31
	500	0.08	0.09	0.75	0.08	0.58
Agent 1	0	0.25	0.25	0.25	0.25	0.00
	50	0.42	0.19	0.20	0.19	0.12
	200	0.65	0.12	0.12	0.11	0.42
	500	0.82	0.06	0.06	0.06	0.71
Agent 2	0	0.25	0.25	0.25	0.25	0.00
	50	0.20	0.35	0.23	0.22	0.06
	200	0.11	0.55	0.18	0.16	0.28
	500	0.05	0.78	0.10	0.07	0.62
Agent 3	0	0.25	0.25	0.25	0.25	0.00
	50	0.21	0.20	0.22	0.37	0.07
	200	0.14	0.13	0.15	0.58	0.30
	500	0.07	0.07	0.09	0.77	0.60

Result: The population has *partitioned* the regime space:

- Agent 1: Bull specialist (SI = 0.71)
- Agent 2: Bear specialist (SI = 0.62)
- Agent 0: Sideways specialist (SI = 0.58)
- Agent 3: Volatile specialist (SI = 0.60)

B.9 The Core Insight: Why $\lambda = 0$ Still Works

The question: If there’s no niche bonus ($\lambda = 0$), why do agents still specialize?

The answer: Competition alone creates path dependence through four mechanisms:

1. **Random initial wins:** In early iterations, winners are determined by noise ϵ . By chance, different agents win in different regimes.
2. **Winner-take-all prevents convergence:** Losers don’t update, so they cannot copy the winner’s strategy.
3. **Method learning creates skill gaps:** Winners accumulate better beliefs about what works in “their” regime. Their Beta distributions narrow around effective methods.
4. **Skill gaps compound:** Once Agent A is slightly better at Bull, they win more Bull rounds, further improving their Bull skills. This is a positive feedback loop.

Table 13: SI comparison: $\lambda = 0$ vs $\lambda = 0.3$. Competition alone ($\lambda = 0$) produces SI > 0.25 , significantly above the random baseline of 0.13.

Domain	$\lambda = 0$ SI	$\lambda = 0.3$ SI	λ Boost
Crypto	0.31	0.75	+142%
Commodities	0.30	0.78	+160%
Weather	0.31	0.71	+129%
Solar	0.26	0.82	+215%
Traffic	0.29	0.68	+134%
Air Quality	0.50	0.80	+60%
Average	0.33	0.76	+140%
vs Random (0.13)	+154%	+485%	—

Conclusion: The niche bonus λ *accelerates* specialization but does not *cause* it. **Competition alone is sufficient.**

B.10 Competition Dynamics: Crowding and Migration

What happens if two agents specialize in the same regime?

Suppose Agents 1 and 6 both become Bull specialists. When a Bull round occurs:

$$\mathbb{E}[\text{Payoff for each Bull specialist}] = \frac{V_{\text{Bull}}}{2} \quad (\text{split the reward}) \quad (33)$$

Meanwhile, Agent 3 (sole Volatile specialist) gets:

$$\mathbb{E}[\text{Payoff in Volatile}] = V_{\text{Volatile}} \quad (\text{no competition}) \quad (34)$$

Result: It’s more profitable to be the sole specialist in an uncrowded niche. Over time, the weaker Bull specialist will “migrate” to Volatile after repeatedly losing to the stronger Bull specialist.

This is **competitive exclusion** from ecology: identical strategies cannot stably coexist.

B.11 Summary: The Complete Picture

Key Takeaways.

1. **Competition is the source:** Specialization emerges from winner-take-all dynamics, not explicit diversity incentives.
2. **No communication needed:** Agents differentiate through individual learning, not coordination.

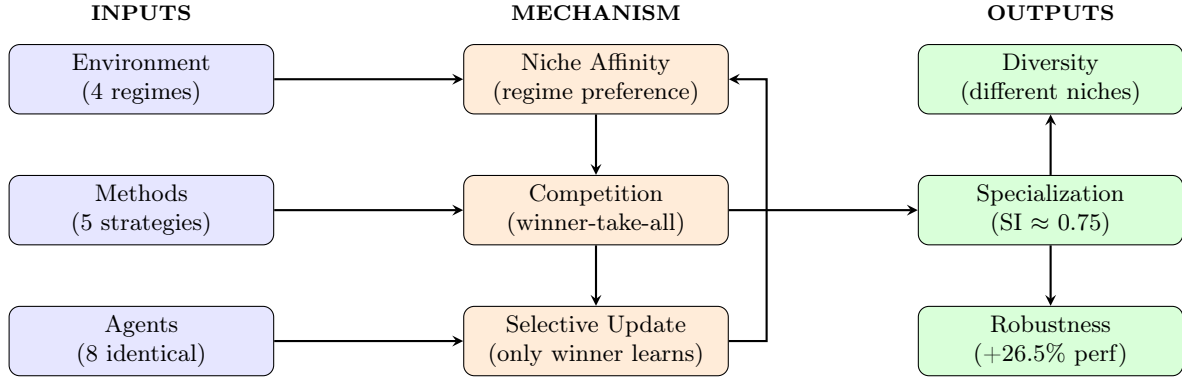


Figure 7: High-level summary: identical agents + competition → emergent specialization.

3. **Ecologically inspired:** The mechanism mirrors competitive exclusion in natural ecosystems.
4. $\lambda = 0$ **still works:** The niche bonus accelerates but does not cause specialization.
5. **Robust and general:** Works across 6 diverse real-world domains with effect sizes $d > 15$.